

Compiling State-Level GSDP Estimates

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PROTOTYPE LEARNING MATERIAL NOTICE

This learning module is compiled from curriculum standards for role-aligned training demonstration.

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1. Introduction

Gross State Domestic Product (GSDP) is the state-level counterpart of GDP. It measures the total value of all final goods and services produced within a state's geographic boundary during a given year. GSDP is compiled by each State's Directorate of Economics and Statistics (DES) following the methodology guidelines issued by MoSPI/CSO.

GSDP estimates are critical for:

- State-level fiscal planning and budgeting
- Determining states' share of Central taxes and grants (Finance Commission)
- Calculating per capita income for inter-state comparisons
- Formulating state economic policies and planning
- Monitoring sustainable development goals at the state level

2. Key Concepts

2.1 GSDP vs. GDP

GSDP follows the same conceptual framework as GDP but is geographically bounded to a state. The sum of GSDPs across all states does not exactly equal national GDP because of statistical discrepancies, treatment of centrally-administered activities, and the problem of allocating multi-location enterprises.

2.2 Net State Domestic Product (NSDP)

$NSDP = GSDP - \text{Consumption of Fixed Capital (depreciation)}$. NSDP at factor cost (excluding subsidies) gives State Domestic Product at factor cost, which approximates income accruing to state residents from state-based production.

2.3 Per Capita GSDP

Per Capita GSDP = GSDP / Mid-year population of state. Used for inter-state comparisons but must be interpreted carefully since it does not capture income inequality within the state.

2.4 District Domestic Product (DDP)

Many states now compile DDP at the district level, extending GSDP methodology downward. This supports district-level planning and identification of lagging regions.

3. Institutional Framework

****Responsibility****: Each State/UT Directorate of Economics and Statistics (DES) is responsible for compiling and publishing GSDP estimates.

****Coordination****: MoSPI/CSO issues Handbook of Instructions on compilation of District Domestic Product and provides methodological guidance for GSDP.

****Publication****: States are expected to publish GSDP estimates annually, but timeliness varies. MoSPI compiles GSDP data from all states and publishes national aggregates.

****Base Year****: States are expected to align with the national base year (currently 2011-12) for comparability. Base year revision requires re-basing the entire series.

4. Sector-wise Methodology for State Accounts

4.1 Agriculture & Allied Activities

State-level crop production data (area under cultivation × yield × state harvest price) compiled by the Directorate of Agriculture. For livestock, data from Livestock Census and state animal husbandry departments. Intermediate consumption ratios applied from national benchmarks or state-specific surveys.

Key challenge: Irrigation and soil conditions vary widely; national IC ratios may not apply uniformly.

4.2 Mining & Quarrying

Data from state geology & mining departments, IBM returns, and royalty collection records. States with major mineral deposits (Chhattisgarh, Jharkhand, Odisha) have well-developed mineral output data. Minor minerals (sand, gravel, stone) are often under-reported.

4.3 Manufacturing

Organised: ASI data disaggregated by state using NIC codes and state codes. Unorganised: State-level NSSO data (often small sample; requires careful use). IIP not available at state level for most states; alternative indicators (power consumption, production indices) used as extrapolators.

4.4 Construction

State-level commodity flow estimates using state-specific cement consumption, steel dispatches, and building permit data (from local bodies). Public sector construction directly estimated from plan expenditure data.

4.5 Services

Trade: State-level trade turnover from commercial tax/GST records. Transport: Vehicle registration data, petrol/diesel sales, and railway data apportioned to states. Financial services: Branch-wise deposit/credit data from RBI allocated to states. Public administration: State budget expenditure data (compensation of employees + intermediate consumption).

5. Allocation of Centrally Administered Sectors

Several activities are centrally administered and cannot be directly allocated to states:

- Railways (allocated by originating traffic/route miles)
- Central government administration (allocated by employee location or population share)
- Post & telecommunications (allocated by subscriber location)
- Defense establishments (not allocated to individual states; counted in national GDP only)
- Central Public Sector Undertakings (allocated by plant/establishment location)

This allocation involves judgment calls and approximations that contribute to discrepancies between the sum of GSDPs and national GDP.

6. Common Challenges in GSDP Compilation

****1. Data availability and timeliness**:** State-level data sources are often less timely than national sources. This leads to longer revision cycles for GSDP.

****2. Multi-location enterprises**:** A firm operating in multiple states poses an allocation problem. Output should be allocated to the state where production occurs, not where the head office is registered.

****3. Base year revision**:** When the national base year changes, states must revise their entire GSDP series. Capacity constraints in state DES offices can delay revision.

****4. Inconsistent methodology**:** Different states may use different IC ratios or extrapolators, reducing inter-state comparability.

****5. Informal sector**:** Informal and household enterprises are particularly difficult to enumerate at the state level given limited NSSO sample sizes.

****6. Price indices**:** State-level price deflators are often unavailable; national WPI/CPI applied uniformly, which may not reflect state-level price dynamics.

7. Quality Assurance Steps

1. ****Internal consistency checks**:** Verify that current-price and constant-price GSDP growth rates are plausible given state economic conditions.

2. ****Cross-sector checks**:** Ensure that agricultural output estimates are consistent with rainfall, cropped area, and procurement data.

3. ****National comparison**:** GSDP growth rates should be broadly consistent with national GDP trends unless state-specific factors justify divergence.

4. ****Revision analysis**:** Document and explain major revisions between estimates. Unexplained large revisions signal data or methodology problems.

5. ****Peer review**:** CSO technical workshops provide a platform for state DES officers to discuss methodology and resolve inter-state comparability issues.

8. Practical Example: Computing Agriculture GVA for a State

Suppose State X has the following data for paddy:

- Area under cultivation: 15 lakh hectares
- Yield: 2.5 tonnes/hectare
- State harvest price: ₹1,800/quintal (= ₹18,000/tonne)

Value of paddy output = $15 \times 2.5 \times 18,000 = ₹16,75,000 \text{ lakh} = ₹16,750 \text{ crore}$

Intermediate consumption ratio (seeds, fertiliser, pesticides, irrigation) = 30% of output

IC = $0.30 \times 6,750 = ₹2,025 \text{ crore}$

GVA from paddy = $6,750 - 2,025 = ₹4,725 \text{ crore}$

Repeat for all crops, livestock, forestry, and fishing; sum to get state Agriculture & Allied GVA.

9. Key Learning Points

1. GSDP follows the same conceptual framework as national GDP but is geographically bounded.
2. State DES offices compile GSDP; MoSPI provides methodological guidance and quality oversight.
3. Centrally-administered sectors require allocation to states using proxy indicators.
4. Multi-location enterprises, informal sector coverage, and limited price data are key challenges.
5. Quality assurance involves internal consistency, cross-sector, and national comparison checks.
6. The sum of GSDPs differs from national GDP due to statistical discrepancies and boundary issues.

10. Learning Outcomes

On completing this material, you should be able to:

- Explain the GSDP framework and its institutional context in India.
- Apply sector-wise methodology to compile state-level GVA estimates.
- Identify and address common challenges in state accounts compilation.
- Describe the treatment of centrally-administered sectors in state accounts.
- Design and conduct quality assurance checks on GSDP estimates.